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Thalassogigantia insuliformis

A Biological Monograph on the Tanayu

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Notice

The present monograph was compiled centuries prior to the cessation of direct Senate administrative oversight of AR-29G and reflects observations conducted while colonial institutions, archival systems, and traditional Tanayu husbandry practices remained intact. Recent covert observation missions, however, have revealed substantial divergence between contemporary conditions on AR-29G and those described herein.

Assessment

Contemporary populations appear to possess little or no institutional memory concerning deliberate offspring redistribution, long-term archipelago management, or other forms of Tanayu husbandry described in this volume. Readers are advised accordingly that the section on *Relations with Human Civilizations* should be interpreted as accounts of documented historical practice.

This assessment should not be considered definitive, as post-oversight protocols prohibit direct inquiry into pre-cessation practices. It is instead derived from long-term covert observation and from the apparent absence of such practices in contemporary AR-29G societies.

Authorized by Directive 14-88,
Office of Abandoned Colony Reassessment

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Contents

1 Introduction **3**

2 Nomenclature **4**

3 Taxonomic Classification **4**

4 General Morphology **5**

5 The Carapace **6**

5.1 Composition and Growth 6
5.2 Erosion and Albasoil Formation 7
5.3 Topography 7

6 Albasoil Layer Regulation **8**

7 Nutrition and Metabolism **9**

7.1 Filter Feeding 9
7.2 Symbiotic Energy Acquisition 9
7.3 Thermoregulation 10

8 Life Cycle **10**

8.1 Reproductive Gatherings 10
8.2 The Tanalet Stage 11
8.3 The Tanaling Stage 12
8.4 Adulthood 13
8.5 Senescence and Death 13

9 Relations with Human Civilizations **14**

9.1 Archipelago Management and Breeding Politics 14
9.2 Diet, Resource Use, and Environmental Management 15

10 Conclusion and Research Limitations **16**

1 Introduction

The organism commonly known as the Tanayu is one of the most remarkable recorded examples of pelagic megafauna capable of sustaining complex terrestrialized ecosystems upon its own body. Adult specimens of *Thalassogigantia insuliformis* resemble mobile islands, bearing exposed calcareous carapaces large enough to support soils, vegetation-like organisms, animal communities, and permanent human settlement.

Figure 1: View of the city-state of [redacted] as observed from low orbit. Image heavily redacted under the Interstellar Post-cessation Non-Interference Agreement.

The present monograph summarizes the current biological understanding of *T. insuliformis*, with particular attention to its taxonomy, external morphology, albasoil-producing physiology, life cycle, symbiotic relations with ranunculan plant forms, and ecological role as a host substrate for human civilizations. Although Tanayu biology has long been recorded in local oral histories, systematic xenobiological study remains difficult. The animals are extremely long-lived, slow-moving, and geographically dispersed, and many of their most important life processes occur over timescales longer than those of ordinary institutional observation.

The Tanayu cannot be understood as merely large marine animals. Their bodies function simultaneously as organisms, landforms, nutrient engines, and civilizational foundations. The distinction between biology, geography, and anthropology is therefore unusually unstable in Tanayu studies. A mature specimen is not only an individual animal, but also a moving island, a primary producer support system, a migratory habitat, and, in many cases, a sovereign or semi-sovereign human homeland.

2 Nomenclature

The Kupina Hala term *Tanayu* is the preferred Interstellar Xenobiology Bureau common name. In Kupina Hala, it is conventionally analyzed as *tana* 'land' and *layu* 'to live; to inhabit', yielding an approximate sense of 'living land' or 'inhabited land'. Some historical and mythological traditions, however, maintain that the name derives instead from *Tanen Yu*, a semi-legendary culture hero associated with the earliest recorded settlements upon the great behemoths. Owing to the absence of reliable documentation, the relationship between these competing etymologies remains unresolved.

3 Taxonomic Classification

Rank	Classification
Family	Thalassogigantidae
Genus	*Thalassogigantia*
Species	*Thalassogigantia insuliformis*

Table 1: Accepted taxonomic placement of the Tanayu.

The genus *Thalassogigantia* contains three commonly cited species. The best attested is *T. insuliformis*, the true Tanayu. [redacted]

Species	Common Name	Status
T. insuliformis	Tanayu	Extant
[redacted]	[redacted]	[redacted]
		[redacted]

Table 2: Recognized and reputed members of *Thalassogigantia*.

4 General Morphology

Adult Tanayu are massive marine organisms combining features superficially reminiscent of turtles, giant manta rays, and horseshoe crabs. This comparison is only descriptive: no close genetic relation to terrestrial fauna is implied. Their dorsal surface is dominated by a continuous calcareous carapace that rises above the waterline in mature individuals. In large adults, the exposed dorsal region may reach several kilometers in length, with some specimens approaching [redacted] from anterior to posterior extremity.

[redacted]

Figure 2: Approximate size comparison between adult *T. insuliformis*, common ocean vessels, and a standard colonial settlement block. Image withheld by order of the Senate.

The living body below the carapace remains mostly submerged. Propulsion is slow and deliberate, achieved by broad lateral swimming structures and subtle body undulations. Tanayu movement is therefore closer, from a human perspective, to the drift of a vessel than to the swimming patterns of ordinary marine animals. Settlements built upon adult specimens may experience their host's migration as a gradual change of climate, currents, and visible stars rather than as perceptible motion.

The anterior surface is adapted for filter feeding. The animal passes enormous quantities of upper-ocean water through specialized feeding structures at the front of the body, extracting plankton-like organisms and other suspended organic matter. Although commonly referred to as filter-feeding organs, they appear to filter for little beyond "not being larger than the mouth", a criterion which, given the size of adult individuals, excludes surprisingly little.

This feeding mode supplies part of the animal's energy needs, but it is insufficient by itself to explain the metabolic requirements of the largest adults. The remaining energy is supplied through extensive symbiotic relationships with photosynthetic organisms living on the carapace.

[redacted] [redacted]

Figure 3: Anterior feeding structures of adult *T. insuliformis*. Left: external view of the cephalic intake region. Right: internal flow pathways during active feeding. Fine anatomical detail withheld by order of the Senate because “encouraging civilians to approach giant mouths is counterproductive.”

5 The Carapace

5.1 Composition and Growth

The defining anatomical feature of *T. insuliformis* is its carapace, a massive structure formed from a continuous solidified calcareous secretion. This material is produced throughout the life of the organism. Unlike the shell of many smaller animals, the Tanayu carapace is not merely protective. It is also an excretory surface, a mineral store, an ecological substrate, and the foundation for the organism’s symbiotic energy system.

The secretion that forms the carapace contains waste products from the animal’s metabolism. Rather than eliminating these products only into the surrounding seawater, the Tanayu converts much of them into hardened calcareous layers. In this sense, the carapace may be understood as a monumental biological waste-processing system. What would be excrement in a smaller animal becomes, in the Tanayu, the basis for soil formation and ecological productivity.

[redacted]

Figure 4: Depiction of the different carapace layers in an adult Tanayu. The top layers are eroded and sand-like, while the lower layers are hardened and unaffected by the elements. Image withheld by order of the Senate.

5.2 Erosion and Albasoil Formation

Once exposed above sea level, the carapace is subjected to wind, rainfall, salt spray, biological abrasion, and thermal stress. Over time, these processes erode the outer surface into sand-like particles. This material is known as *albasoil*.

Albasoil is rich in biologically available minerals and organic residues. It functions as a natural fertilizer, supporting the growth of specialized plant-like organisms on the dorsal surface. The continual erosion of older carapace and deposition of new calcareous material creates a slow but stable nutrient cycle.

5.3 Topography

The carapace is not uniformly flat. Mature Tanayu commonly develop ridges, basins, fissures, terraces, and elevated spines. These features arise from uneven secretion, differential erosion, mechanical stress, and biological colonization. The resulting landscape may include albasoil plains, rain-fed basins, cliff-like margins, and high dorsal crests rising hundreds of meters above the surrounding ocean.

Figure 5: Representative dorsal landscapes of mature *T. insuliformis*. Left: albasoil basin supporting intensive agriculture. Right: elevated dorsal ridge system on [redacted]. Exact locations withheld under the Post-cessation Non-Interference Agreement.

Human settlement patterns are strongly shaped by this topography. Stable basins collect water and albasoil and are preferred for agriculture. Elevated ridges are used for observation, ritual, and defense. Marginal slopes are often dangerous because of landslides that could throw one into the ocean, but they are the only place to dock vessels.

6 Albasoil Layer Regulation

The Tanayu does not produce carapace material at a constant rate throughout life. As individuals reach advanced size and age, their growth and excretion rates appear to diminish. This regulated growth prevents uncontrolled accumulation of albasoil and reduces the risk of ecological imbalance on the dorsal surface.

The mechanism governing this regulation remains incompletely understood. It is likely to involve both internal metabolic signals and external feedback from the carapace ecosystem. A heavily vegetated, nutrient-stable carapace may require less new secretion than a damaged or recently eroded surface.

Field observations suggest that Tanayu can influence where new calcareous waste is deposited. Rather than secreting uniformly across the entire dorsal surface, individuals appear capable of concentrating deposition in regions where erosion has exposed older layers or where ecological productivity has declined.

[redacted]

Figure 6: Observed dorsal secretion patterns in a mature individual over a fifty-year interval. Regional deposition rates and associated settlement locations redacted under the Post-cessation Non-Interference Agreement.

This strategic excretion control is central to the long-term stability of Tanayu ecosystems. By placing new material where it can be most effectively incorporated into albasoil, the animal minimizes waste, reinforces weakened carapace zones, and maintains the nutrient base required by its symbionts.

When dorsal deposition is unnecessary or potentially harmful, Tanayu can excrete through lateral pores submerged beneath the waterline. These pores discharge calcareous and nutrient-rich waste directly into the surrounding sea. Lateral secretion prevents excessive carapace growth and distributes nutrients to nearby marine ecosystems. For most host civilizations, the process is known only indirectly through the increased biological activity observed in the waters surrounding the Tanayu.

7 Nutrition and Metabolism

7.1 Filter Feeding

Tanayu feed primarily by filtering [redacted] organisms from the upper ocean. Their routes tend to follow regions of high surface productivity, especially warm currents, seasonal bloom zones, and areas enriched by lateral secretion from other individuals. Because adults move slowly, migration routes must be metabolically predictable over long periods.

Filter feeding provides proteins, lipids, trace minerals, and other compounds difficult to obtain from photosynthetic symbiosis alone. Juveniles, whose carapaces have not yet developed extensive terrestrial ecosystems, depend more heavily on filter feeding than adults.

7.2 Symbiotic Energy Acquisition

The immense size of adult Tanayu cannot be sustained by filter feeding alone. Much of their energy is obtained indirectly from photosynthetic organisms living on the carapace. These organisms produce sugars and other organic compounds using sunlight, water, and atmospheric carbon dioxide. Through specialized conduits and root-like interfaces, a portion of this surplus is transferred into the Tanayu's body.

This arrangement makes the Tanayu a partially solar-powered animal through symbiosis. The animal provides substrate, minerals, moisture gradients, and protection; the plants provide energy-rich compounds and assist in thermal regulation. The result is a composite biological system whose productivity exceeds that of either partner in isolation.

[redacted]

Figure 7: Depiction of albasoil generation and algae root conduits forming the basis of the symbiotic system on a Tanayu's carapace. Image withheld by order of the Senate.

Tanayu are not continuously active. Along their migratory routes, individuals regularly pause at traditional resting sites where conditions for photosynthesis are particularly favorable. During these periods, which may last from several days to several weeks, the animals remain almost completely stationary, drifting only slightly with currents and tides.

These resting phases substantially reduce energetic expenditure while maximizing energy intake from their photosynthetic symbionts. Host civilizations often structure agricultural, religious, and commercial activities around these predictable pauses, treating them as seasonal events. The location and duration of resting periods vary among individuals and migratory traditions, but they are considered an essential component of Tanayu metabolism.

7.3 Thermoregulation

The vegetated carapace contributes to temperature regulation. Photosynthetic cover reduces excessive heating of exposed calcareous surfaces, while transpiration-like processes help moderate local humidity and surface temperature. These effects influence the Tanayu's metabolism, especially in shallow or equatorial waters where overheating of exposed dorsal tissue could become dangerous.

Human inhabitants also benefit from this regulation. Regions with healthy plant cover are cooler, more humid, and more habitable than bare albasoil fields. Deforestation or overharvesting of symbiotic vegetation can therefore harm both the Tanayu and its human residents.

8 Life Cycle

8.1 Reproductive Gatherings

Tanayu reproduction occurs at specialized breeding grounds where virtually all mature individuals converge every few centuries. These immense gatherings represent one of the most significant biological events on AR-29G and are central not only to Tanayu ecology but also to the political life of many host civilizations.

The triggers for these migrations remain poorly understood. Long-period climatic cycles, inherited migratory memory, ocean chemistry, and social cues have all been proposed as contributing factors.

Tanayu are simultaneous hermaphrodites. Every individual is capable of both laying eggs and fertilizing those of others, an advantageous strategy for organisms that encounter conspecifics only rarely.

Eggs are deposited in sheltered waters associated with the breeding grounds. Embryonic development is rapid relative to the species' extraordinary lifespan, and hatching occurs after only a few weeks.

8.2 The Tanalet Stage

Newly hatched Tanayu, known as *tanalets*, are approximately the size of a human hand and are fully aquatic.

Shortly after hatching, tanalets instinctively seek the protection of a nearby adult, usually identifying the nearest individual as their mother regardless of genetic parentage. Once an attachment has been established, the young remain beneath or immediately behind their chosen adult, sheltering within the hydrodynamic wake produced by the larger animal.

Figure 8: Maternal associations in early developmental stages. Left: newly attached tanalets occupying the hydrodynamic shelter zone beneath an adult. Right: mixed-age nursery aggregation at a reproductive gathering. Images withheld by order of the Senate.

This maternal association defines the tanalet stage. Adults make no apparent distinction between genetic offspring and adopted young, and entire cohorts may become attached to individuals unrelated to them.

The submerged carapace of tanalets is colonized during this period by pioneer symbionts known as tadpole algae. These organisms attach to the developing carapace and remain dormant or minimally active while the host remains submerged.

Because their dorsal symbiosis is not yet functional, tanalets depend almost entirely on filter feeding for energy. They consume plankton-like organisms in surface waters and frequently compete with siblings for food resources disturbed or left behind by the adult they follow.

Mortality during this stage is extremely high. Predation, starvation, disease, and competition reduce survival dramatically. Of the thousands of hatchlings produced during a single reproductive event, typically only three to five survive long enough to reach the next stage. Human intervention, however, can often increase survival rates substantially.

8.3 The Tanaling Stage

After approximately a hundred years, a surviving tanalet reaches roughly one eighth of the size of the adult it follows. At this point, the juvenile rises to the surface and permanently exposes portions of its dorsal carapace above the waterline. Individuals in this developmental phase are known as *tanalings*.

The emergence of the carapace initiates profound physiological changes. Dormant tadpole algae transition from their aquatic form to terrestrialized growth and begin spreading across the exposed albasoil surface as extensive clonal colonies. Through this symbiosis, tanalings gain access to photosynthetically derived energy for the first time.

Their increased size also allows a broader diet. In addition to filtering plankton-like organisms, tanalings are capable of consuming fish and other comparatively large marine animals.

Figure 9: Representative tanaling behavior. Left: recently surfaced tanaling exhibiting early dorsal colonization by tadpole algae. Right: sibling formation during long-distance migration. Image withheld by order of the Senate.

Socially, tanalings continue to accompany the adult they have followed since hatching. They participate in migrations and behave similarly to adults in most respects, but they remain reproductively and socially subordinate because they have not yet acquired offspring of their own.

Tanalings commonly travel behind their mother alongside numerous siblings. Competition for food within these groups often produces distinctive migratory formations resembling the V-shaped arrangements observed in migrating birds on other planets. Such spacing allows each individual to exploit a different portion of the surrounding ocean while still maintaining cohesion with the group.

Throughout this stage, growth continues steadily. Larger tanalings are generally more successful at attracting or acquiring tanalets during subsequent breeding seasons.

8.4 Adulthood

Acquisition of offspring rather than simple age or size appears to be the defining criterion of social and biological maturity in *T. insuliformis*. A tanaling that successfully acquires one or more tanalets in a reproductive gathering enters adulthood. Upon becoming an adult, the individual leaves its mother's group and establishes an independent migratory route.

Competition for migratory territories is intense. Newly mature adults must rapidly secure productive feeding paths and favorable resting sites. Some individuals form long-lasting associations and migrate cooperatively, while others remain solitary.

The departure of newly mature adults naturally limits the long-term size of Tanayu groups. Consequently, stable multi-generational archipelagos are rare in unmanaged populations.

8.5 Senescence and Death

Tanayu may survive for several millennia. Upon death, the body remains afloat for a variable period before eventually sinking into the deep ocean.

If the deceased individual was the primary adult of an archipelago, succession depends upon the presence or absence of dependent tanalets.

When dependent tanalets remain, they disperse among the surviving tanalings. Those tanalings that acquire offspring immediately enter adulthood and typically depart to establish independent migratory routes. Individuals that fail to obtain offspring instead continue following successful siblings.

If no dependent tanalets remain at the time of death, the oldest tanaling ordinarily assumes the role of principal adult, with younger siblings continuing to follow it until future reproductive gatherings allow them to acquire offspring of their own.

Figure 10: Observed terminal sequence in an elderly *T. insuliformis*. Final resting location, associated settlements, and succession records withheld at the request of descendant communities.

9 Relations with Human Civilizations

9.1 Archipelago Management and Breeding Politics

Tanayu naturally organize themselves into social groups commonly referred to as archipelagos. Typically, a mature adult occupies the leading position within such a group and determines its migratory route, while dependent tanalings and tanalets follow.

In unmanaged populations, archipelagos tend to fragment over time. Once a tanaling acquires offspring and reaches adulthood, it normally leaves its mother's group to establish an independent migratory path. This continual fission limits the long-term size and persistence of naturally occurring archipelagos.

Human societies have profoundly altered this process. During reproductive gatherings, host civilizations frequently attempt to secure additional offspring, both to increase the size of their populations and to preserve the cohesion of their archipelagos. Competition for tanalets can become intense, and some breeding grounds have historically been the sites of large-scale conflicts involving specialized naval and underwater combat forces.

[redacted]

Figure 11: Distribution of managed archipelagos and major reproductive gathering sites. Breeding grounds, naval installations, and political boundaries redacted under the Post-cessation Non-Interference Agreement.

Many societies deliberately redistribute offspring among the Tanayu under their stewardship. Newly acquired tanalets may be placed under the care of established adults rather than their biological or adoptive parents, encouraging mature tanalings to remain attached to the principal Tanayu instead of departing to found new archipelagos. Similar interventions may occur following the death of a principal adult, when dependent tanalets are redistributed among selected tanalings to preserve political continuity and prevent fragmentation. Through such practices, humans effectively manage the social structure of their host organisms.

As a consequence, many of the great archipelagos inhabited today are not entirely natural social formations, but partially anthropogenic constructs maintained through centuries of careful intervention.

9.2 Diet, Resource Use, and Environmental Management

Human diets on Tanayu commonly include harvested ranunculan tissues, cultivated crops grown in albasoil, marine animals, preserved planktonic products, and water-rich plants such as [redacted]. Tadpole algae and related organisms may be eaten directly or processed into nutrient-dense pastes.

[redacted] [redacted]

Figure 12: Representative food production systems on managed Tanayu. Left: cultivated albasoil terraces supporting mixed agriculture. Right: processing of harvested tadpole algae into preserved food products. Image withheld by order of the Senate for worker-privacy reasons.

Resource use depends heavily on sustainable harvest. Because the dorsal ecosystem also nourishes the Tanayu, excessive consumption of symbiotic plants can indirectly weaken or starve the host. Agriculture, construction, quarrying of albasoil, deforestation, and waste disposal may likewise alter the health of the dorsal ecosystem. Consequently, most host societies maintain strict ecological regulations governing harvest cycles, land use, settlement placement, and the treatment of secretion zones.

Long-term occupation has produced sophisticated ecosystem-management practices throughout AR-29G. Rotational harvest systems, protected groves, seasonal restrictions, and ritual prohibitions are common. In well-managed regions, human cultivation may even increase local biodiversity through the creation of ponds, terraces, compost fields, and managed woodlands. Poor management, however, can result in erosion, nutrient depletion, and exposure of the underlying carapace.

Host societies also possess extensive ecological knowledge of Tanayu migration. Generations of observation allow communities to anticipate changing marine resources, ocean-floor nutrients, fisheries, and climatic conditions encountered along migratory routes. In many regions, this knowledge forms the basis of agricultural calendars, navigation, trade, and religious practice.

10 Conclusion and Research Limitations

Current knowledge of *Thalassogigantia insuliformis* remains constrained by several factors. The species' extraordinary longevity makes direct observation of complete life cycles exceedingly rare, while many host communities restrict access to biologically sensitive regions of the carapace. In addition, some of the most important physiological structures lie beneath kilometers of living tissue and calcareous material, rendering invasive study both ethically and practically unacceptable.

The extensive influence of human civilizations presents an additional challenge. Human activity has shaped many dorsal ecosystems for centuries or millennia, making it difficult to distinguish unmanaged biological processes from landscapes and social systems produced through long-term stewardship.

Future research should therefore proceed in close cooperation with host civilizations and with due consideration for the exceptional ecological and cultural significance of the species.

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